

1 Appendix A

2 **Table 1. Complete List of Orthogonal Visual Attribute Dictionary for 15 Categories of Concrete Defects**

Defect Name	Official ACI/GB Standard Definition	Core Attributes (W_3)	Auxiliary Attributes (W_1)
Disintegration	Reduction into small fragments and subsequently into particles.	Surface_Attachments: Granular Material_Exposure: Exposed_Coarse_Particles	Edge_Condition: Broken Local_Profile: Depression
Spalling	A fragment, usually in the shape of a flake, detached from a larger mass by a blow, the action of weather, pressure, or expansion within the larger mass.	Local_Profile: Depression Material_Exposure: Exposed_Coarse_Particles	Edge_Condition: Sharp Surface_Texture: Rough
Scaling	Local flaking or peeling away of the near-surface portion of hardened concrete or mortar.	Material_Exposure: Exposed_Coarse_Particles Local_Profile: Layered_Flake	Surface_Texture: Rough Edge_Condition: Broken
Cold Joint	A joint or discontinuity resulting from a delay in placement of sufficient duration to preclude intermingling and bonding of the material, or where mortar or plaster rejoin or meet.	Relative_Position: Joint Global_Shape: Linear	Path_Regularity: Straight Local_Profile: Depression
Bughole	Small regular or irregular cavities, usually not exceeding 15 mm in diameter, resulting from entrapment of air bubbles in the surface of formed concrete.	Material_Exposure: No_Exposure Local_Profile: Small_Pits	Porosity: High_Porosity Global_Shape: Circular Edge_Condition: Sharp
Honeycomb	Voids left in concrete between coarse aggregates due to inadequate consolidation.	Material_Exposure: Exposed_Coarse_Particles Porosity: High_Porosity	Surface_Texture: Honeycombed Local_Profile: Deep_Depression
Oriented Cracking	A complete or incomplete separation of either concrete or masonry into two or more parts produced by breaking or fracturing.	Global_Shape: Linear Branching_Pattern: None	Path_Regularity: Meandering
Mapped Cracking	Intersecting cracks that extend below the surface of hardened concrete that vary in width from fine and barely visible to open and well-defined.	Global_Shape: Reticulate Branching_Pattern: Interconnected_Network	Path_Regularity: Meandering
Exposed Rebar	The phenomenon where steel bars inside a structural component are not covered by concrete and are directly exposed.(The original text is in Chinese)	Material_Exposure: Exposed_Metal_Bar	Local_Profile: Depression
Efflorescence	A generally white deposit formed when water-soluble compounds emerge in solution from concrete, masonry, or	Surface_Color: White Surface_Attachments: Crystalline	Edge_Condition: Diffuse

Defect Name	Official ACI/GB Standard Definition	Core Attributes (W_3)	Auxiliary Attributes (W_1)
Rust Stain	plaster substrates and precipitate by reaction such as carbonation or crystallize by evaporation. Reddish-brown or yellowish-brown stains deposited on the concrete surface caused by the corrosion of internal metal/rebar, where iron oxides leach out with moisture.(The original text is in Chinese)	Surface_Color: Reddish_Brown	Stain_Pattern: Dripping
Water Stain	Noticeable boundary outlines and color changes left by moisture evaporation on the concrete surface, without obvious water droplets falling. (The original text is in Chinese)	Surface_Color: Brown/Dark Edge_Condition: Sharp	Global_Shape: Irregular Surface_Orientation: Ceiling_Surface
Stalactite	A downward-pointing deposit formed as an accretion of mineral matter produced by evaporation of dripping water from the surface of concrete, commonly shaped like an icicle.	Surface_Attachments: Solidified_Drip Surface_Orientation: Ceiling_Surface	Surface_Color: White Local_Profile: Icicle_Shape
Crusting	Crusting occurs when the top surface of the concrete begins to set or dry out rapidly while the underlying concrete is still in a plastic state.	Local_Profile: Brittle_Flake Material_Exposure: Exposed_Cement_Paste	Edge_Condition: Broken
Biological Stain	Colonization of damp concrete surfaces by living organisms such as algae, fungi, moss, or lichen, typically presenting as green, black, or brown pigmented patches or fuzzy deposits.	Surface_Color: Green Surface_Attachments: Fuzzy_Patch	Stain_Pattern: Blotchy